



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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UNIT TEST-1, JULY - 2024

Time Allowed : 1.30 Hours]

MATHEMATICS

[Max. Marks : 50

Part - I

1. Answer all the questions by choosing the correct answer from the given 4 alternatives.
2. Write question number, correct option and corresponding answer
3. Each question carries 1 mark. 10x1=10

1. If $A = \{(x, y) : y = \sin x, x \in \mathbb{R}\}$ and $B = \{(x, y) : y = \cos x, x \in \mathbb{R}\}$ then $A \cap B$ contains
 - (a) no element
 - (b) infinitely many elements
 - (c) only one element
 - (d) cannot be determined
2. Let A and B be subsets of the universal set N , the set of natural numbers. Then $A \cup [(A \cap B) \cup B]$ is
 - (a) A
 - (b) A^c
 - (c) B
 - (d) N
3. If two sets A and B have 17 elements in common, then the number of elements common to the set $A \times B$ and $B \times A$ is
 - (a) 2^{17}
 - (b) 17^2
 - (c) 34
 - (d) insufficient data
4. Let R be the universal relation on a set X with more than one element. Then R is
 - (a) not reflexive
 - (b) non symmetric
 - (c) transitive
 - (d) none of the above
5. The solution of $5x-1 < 24$ and $5x+1 > -24$ is
 - (a) (4,5)
 - (b) (-5,-4)
 - (c) (-5,5)
 - (d) (-5,4)
6. The solution set of the following inequality $|x-1| \geq |x-3|$ is
 - (a) $[0,2]$
 - (b) $[2, \infty)$
 - (c) $(0,2)$
 - (d) $(-\infty, 2)$
7. If 3 is the logarithm of 343, then the base is
 - (a) 5
 - (b) 7
 - (c) 6
 - (d) 9
8. If a and b are the roots of the equation $x^2 - kx + 16 = 0$ and satisfy $a^2 + b^2 = 32$, then the value of k is
 - (a) 10
 - (b) -8
 - (c) -8, 8
 - (d) 6
9. No of reflexive relations on a set containing n elements
 - (a) 2^{n^2}
 - (b) 2^{n^2-n}
 - (c) 2^{2n}
 - (d) $2^{\frac{n^2+n}{2}}$
10. Range of the function $f(x) = \begin{cases} \frac{x}{|x|} & x \neq 0 \\ 0 & x = 0 \end{cases}$ is
 - (a) $(-\infty, \infty)$
 - (b) $(-1,1)$
 - (c) $\{-1,0,1\}$
 - (d) $(0, \infty)$

PART - II

1. Answer any four of the following 2. Each question carry 2 marks.

4x2=8

3. Question number 16 is compulsory.

11. If $n(P(A)) = 1024$, $n(A \cup B) = 15$ and $n(P(B)) = 32$, then find $n(A \cap B)$.
12. Discuss the following relations for reflexivity, symmetricity and transitivity: On the set of natural numbers the relation R defined by " xRy if $x + 2y = 1$ ".



13. Solve: $-x^2 + 3x - 2 \geq 0$.

14. Find the number of solutions of $x^2 + |x-1| = 1$.

15. Find the logarithm of 1728 to the base $2\sqrt{3}$.

16. Find the domain of $f(x) = \frac{1}{1 - 2 \cos x}$.

PART - III

1. Answer any four of the following. 2. Each question carry 3 marks.

4x3=12

3. Question number 22 is compulsory.

17. Simplify: $(3^{-4})^{1/3}$

18. Solve $\frac{x+1}{x+3} < 3$.

19. Solve $\frac{1}{|2x-1|} < 6$ and express the solution using the interval notation.

20. Write the steps to obtain the graph of the function $y = 3(x-1)^2 + 5$ from the graph $y = x^2$.

21. Prove that $((A \cup B \cup C) \cap (A \cap B \cap C)) \cup ((A \cup B \cup C) \cap (B \cap C)) = B \cap C$.

22. In the set Z of integers, define mRn if $m-n$ is a multiple of 12. Prove that R is an equivalence relation.

PART - IV

1. Answer all the questions. 2. Each question carry 5 marks.

4x5=20

23. a) In a survey of 5000 persons in a town, it was found that 45% of the persons know Language A, 25% know Language B, 10% know Language C, 5% know Languages A and B, 4% know Languages B and C, and 4% know Languages A and C. If 3% of the persons know all the three Languages, find the number of persons who knows only Language A.

(OR)

- b) Let P be the set of all triangles in a plane and R be the relation defined on P as aRb if a is similar to b. Prove that R is an equivalence relation.

24. a) If $f: R \rightarrow R$ is defined by $f(x) = 2x - 3$ prove that f is a bijection and find its inverse.

(OR)

- b) From the curve $y = |x|$, draw (i) $y = |x - 1| + 1$ (ii) $y = |x + 1| - 1$ (iii) $y = |x + 2| - 3$.

25. a) A manufacturer has 600 litres of a 12 percent solution of acid. How many litres of a 30 percent acid solution must be added to it so that the acid content in the resulting mixture will be more than 15 percent but less than 18 percent?

(OR)

- b) If one root of $k(x - 1)^2 = 5x - 7$ is double the other root, show that $k = 2$ or -25 .

26. a) Resolve into partial fractions: $\frac{2x}{(x^2+1)(x-1)}$

(OR)

- b) Solve $\log_{6x} (x^2 - 6x + 65) = 2$.

